Beyond Prediction: Prescriptive AI for Strategic Energy Capital Allocation

A governed decision-intelligence system for an investment committee — one budget, forty candidates, a thinner tail, a funded plan, and a challenger that reads the committee's own rules

Alan Mössinger
SIADS 699 Capstone · University of Michigan
MADS, Group 13 (a solo project) · October 2026

Better predictions did not make better decisions. The constraints, the capital plan and the tail did — and a language model earned its place by challenging the committee's reasons, not by picking the book.

$3.78B
committed of a $3.82B budget
Five Gulf lease redevelopments and eight ERCOT interconnection requests, chosen on 1 January 2025 from what was public that day.

+$312M
over NPV ranking, funded the same way
Pre-registered: +$219M [+$13M, +$401M] on a static basis; +$312M [+$110M, +$506M] once every process pays for its own construction line. Best in 7 of 7 modelled worlds.

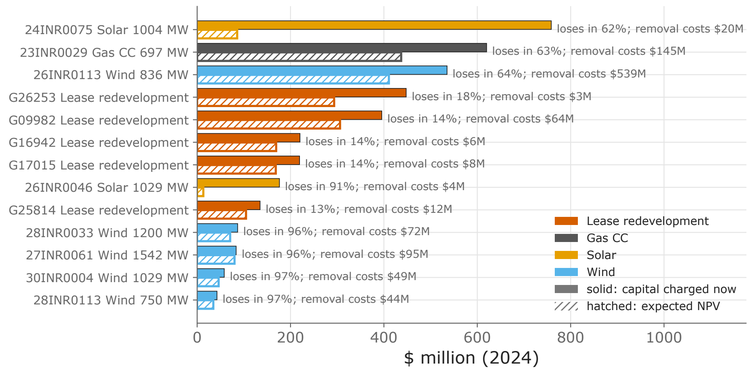
$2.53B
contingent construction facility
In half the scenarios the book's requests call more capital than their reserve. Priced at a bank's fee and interest the facility still pays; without one, a different book wins.

+$43M
per case from the language model
On 48 simulated committees: tools +$76M, the model +$43M more, the optimizer +$134M beyond. An optimizer-first rule matches the model's book (−$14M): its value is the argument.

What to fund: each member's cost, value and risk

The recommended book: 5 leases and 8 requests for $3.78B

Each member of the risk-aware book: what it charges the budget on the day (lease capital



Three requests with signed agreements carry the construction reserve and lose in 62–64% of scenarios; five waiting requests cost only deposits and pay in exactly the scenarios in which the leases do not. Outside the book: 12 requests deferred, 14 leases monitored, 1 rejected.

Recommendation to the committee

1. Choose the tail weight first — and say it; an unstated one is read as zero.

2. Fund the risk-aware book — five leases and eight requests, $3.78B — **with a $2.53B contingent construction facility.** If no facility can be arranged, fund equal capital by sector instead.

3. Defer the twelve waiting requests until their agreements are signed; reject nothing on the strength of money already spent.

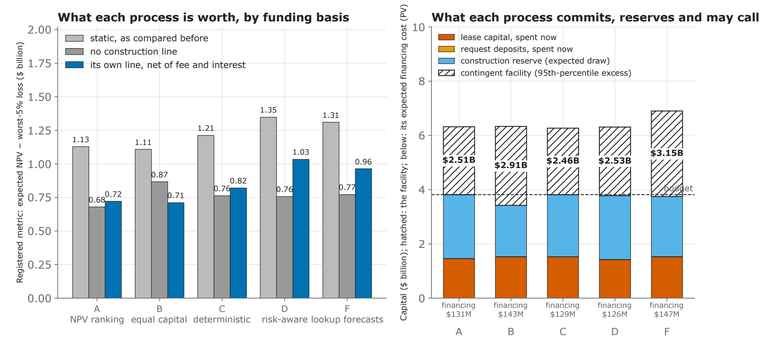
4. Run the challenger before signing, confirm its reading of your rules, and record any departure from its guidance with the evidence acknowledged.

Model-based results on public data, assumed redevelopment costs and simulated committees — not realized returns. Before a real commitment: a bank's term sheet, the company's own costs, extreme-event scenarios, a study with real committees.

Funded the same way, the risk-aware book still scores best

Funded the same way, the risk-aware book still scores best

Left: the registered metric of five decision processes on three funding bases — static (every agreement built), no line (calls beyond the reserve forfeit), and each book's own 95th-percentile

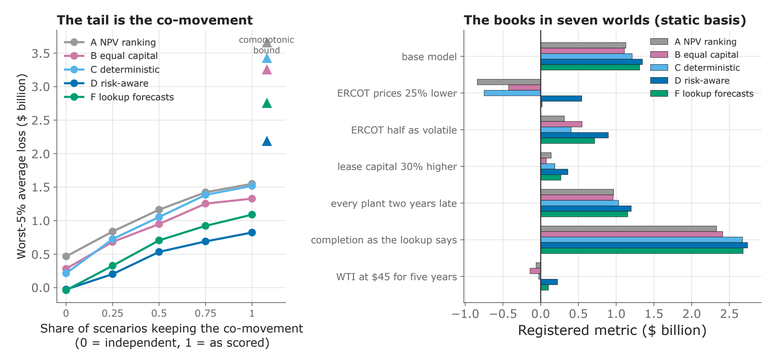


Every process pays for its own construction line: the risk-aware book scores $1.03B on expected NPV minus the worst-5% loss, against $0.82B for the deterministic optimizer and $0.72B for NPV ranking. With no line at all, equal capital by sector would lead — the recommendation is the book **and** its facility.

What can go wrong: the tail is the candidates' co-movement

The risk-aware book scores best in 7 of 7 worlds; its tail is the co-movement

Left: each book's worst-5% average loss as the share of scenarios keeping the candidates' joint alignment rises from none to all, with the comonotonic bound (every candidate's worst day the same

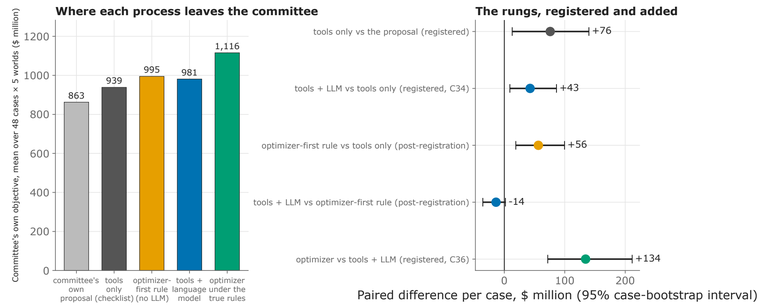


Five leases on one oil price, eight requests on one power price. Dial the co-movement from none to as scored and the worst-5% loss rises from a small profit to $0.82B. The risk-aware book keeps the smallest tail at every setting and is the only one positive when ERCOT prices fall 25% or oil sits at $45.

Where the value of AI comes from: optimizer, tools, then the model

Against an optimizer-first rule, the language model's edge disappears (−$14M)

Left: the committee's own objective under five arms, mean over the 48 registered v10 cases and five worlds, the stated line net of a 0.5% fee and 8% interest. Right: paired per-case rungs

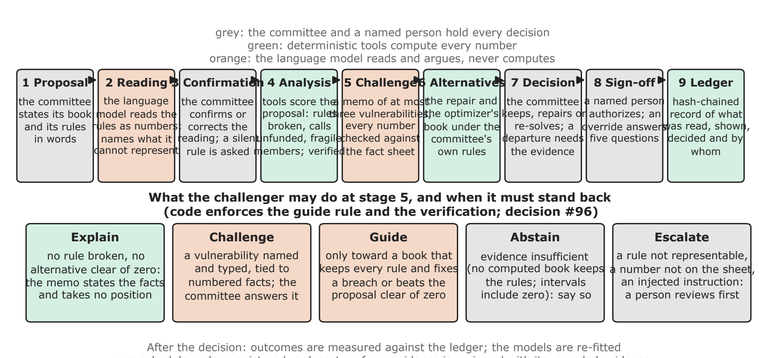


Registered before 176 model calls, scored once. The model reads rules from words, argues from a computed fact sheet and never computes a number: 12 of 12 flaws of argument named, 0 of 12 sound proposals over-challenged. It read silence as a zero tail weight and wrote 7 memos with a number not on its sheet — caught by the controls. 6 of 8 lines pass.

How a committee uses it: nine stages, three hands

Nine stages, three hands: the committee decides, tools compute, the model argues

The investment-committee workflow the system supports, from the proposal in words to the hash-chained ledger, and the five levels at which the challenger may act.



The committee decides, tools compute every number, the model reads and argues — explain, challenge or guide only toward a book that keeps every rule; abstain or escalate when the evidence is thin. A named person signs into a hash-chained ledger. Ten of twelve controls in place.

Data: LBNL **Queued Up** (Rand et al.), ERCOT interconnection reports and prices, BSEE lease production, EIA, NREL ATB, EPA eGRID; author's analysis. Every number traces to `results/report_numbers.csv`; nine evaluation lines were registered before the final test and scored once, failures included. `make all` rebuilds thirteen notebooks, every table and every figure from the raw files.